

FISTER SIGMUNDSTAD, Norway

WMO: 014220

Lat: 59.1597N

Lon: 6.0364E

Elev: 82

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	22.5	5.3	7.3	24.2	9.1	8.8	25.1	25.6	41.0	21.6	41.1	6.4	100	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	76.2	62.3	72.3	60.9	69.1	59.5	64.4	72.1	62.6	69.8	60.9	67.1	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	80.6	67.4	59.4	75.8	65.8	57.9	71.9	64.3	29.6	71.8	28.3	69.8	27.1	67.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.4	14.0	19.8	82.9	4.3	4.5	16.7	86.2	14.2	88.8	11.8	91.4	8.8	94.7		
(5)			17.1	66.2	4.0	2.5	14.2	68.1	11.9	69.5	9.6	71.0	6.7	72.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	35.6	36.3	39.6	45.2	51.6	56.2	61.0	60.0	55.8	48.5	42.2	38.3		
	DBStd	10.43	6.26	5.93	4.67	5.33	6.52	4.57	5.15	3.94	4.29	4.96	6.30	6.72		
	HDD50	2066	447	385	323	164	53	3	0	0	5	85	239	362		
	HDD65	6401	912	805	788	593	419	269	151	165	278	511	684	826		
	CDD50	1187	0	0	0	20	104	189	340	310	179	39	5	1		
	CDD65	47	0	0	0	0	5	5	26	9	2	0	0	0		
	CDH74	222	0	0	0	0	19	25	144	31	3	0	0	0		
	CDH80	34	0	0	0	0	2	2	27	3	0	0	0	0		
	WSAvg	6.1	7.1	6.2	6.2	5.8	5.8	5.9	5.0	5.5	5.7	6.3	6.5	7.4		
	PrecAvg	69.4	7.3	6.1	5.4	3.7	3.2	3.7	4.6	5.8	6.9	8.0	7.6	7.8		
(15) Precipitation	PrecMax	91.3	14.8	14.8	12.0	6.2	8.0	6.5	10.2	10.6	12.6	16.0	14.7	14.6		
	PrecMin	47.4	0.0	0.0	0.2	1.6	0.2	0.7	2.0	1.4	2.1	2.1	0.9	0.7		
	PrecStd	11.3	4.0	3.9	2.8	1.3	1.8	1.8	2.0	2.1	3.1	3.2	4.0	4.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	49.9	53.2	65.4	76.2	77.8	83.6	78.3	72.3	65.1	58.4	52.2		
		MCWB	44.7	44.1	46.2	50.5	59.9	59.8	64.8	63.5	60.2	56.7	51.5	48.1		
	2%	DB	46.8	46.9	49.4	60.9	71.2	71.4	78.1	72.6	67.0	60.6	54.3	50.1		
		MCWB	44.0	43.5	44.3	49.4	57.6	60.0	63.6	60.7	57.9	54.5	48.8	47.4		
	5%	DB	45.3	45.5	47.0	57.8	66.4	66.8	73.4	69.3	65.0	57.4	52.0	48.3		
		MCWB	43.0	42.6	42.6	48.2	55.7	57.8	62.9	59.8	57.1	51.9	47.7	45.6		
	10%	DB	43.8	44.1	45.6	53.7	62.3	63.7	69.5	66.7	62.7	55.2	50.3	46.7		
		MCWB	41.3	41.4	41.8	46.6	53.0	56.1	61.3	58.8	56.0	50.3	46.2	43.6		
	0.4%	WB	46.1	45.9	47.7	52.2	61.9	63.3	67.9	65.7	62.9	58.8	52.5	49.5		
		MCDB	47.3	47.5	50.1	60.7	73.3	72.3	78.9	74.2	70.1	62.6	56.3	50.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.8	44.6	45.4	50.4	58.8	60.6	65.4	63.2	60.3	55.3	50.4	47.8		
		MCDB	46.0	46.3	47.7	58.1	68.4	68.1	74.7	69.4	64.8	59.1	53.3	49.4		
	5%	WB	43.4	43.4	44.1	49.1	56.2	58.6	63.7	61.2	58.3	53.6	48.8	46.1		
		MCDB	44.9	45.2	46.4	56.0	64.4	65.0	71.0	66.4	63.2	56.6	51.6	48.0		
	10%	WB	41.5	41.9	42.9	47.5	54.1	57.0	61.9	59.8	56.6	51.3	46.7	44.1		
		MCDB	43.6	44.0	45.0	53.1	61.0	62.8	68.9	64.9	61.2	54.2	49.6	46.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.3	8.4	11.1	12.1	11.4	11.2	10.7	9.8	8.9	7.2	6.8		
		MCDBR	7.8	7.4	9.2	16.5	18.7	17.1	18.0	14.7	13.0	10.8	8.0	7.5		
		MCWBR	7.2	6.4	6.6	8.8	9.5	9.0	8.2	7.4	7.1	7.3	6.4	7.2		
		MCDBR	7.5	7.5	8.4	14.1	17.3	14.6	15.7	12.3	11.6	8.8	7.8	7.6		
	5% WB	MCWBR	7.5	6.9	6.8	8.2	9.5	8.4	8.0	7.4	7.4	7.2	7.5	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.309	0.335	0.369	0.364	0.358	0.373	0.370	0.349	0.329	0.292	0.259		
	taud		2.166	2.259	2.281	2.271	2.398	2.457	2.434	2.437	2.492	2.448	2.271	2.104		
	Ebn at Noon		163	223	253	264	275	278	271	262	250	217	161	126		
	Edh at Noon		15	23	30	35	34	32	32	30	24	19	13	10		
(44) All-Sky Solar Radiation	RadAvg		98	290	687	1227	1573	1705	1512	1170	750	364	132	64		
	RadStd		19	57	107	163	213	197	172	102	73	54	22	9		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A	

Nomenclature: See separate page